

AI Use Declaration Form

Course: Introduction to Artificial Intelligence

Lab Title: Lab 1 Part A — Uninformed Search Algorithms

Student Name: Fatma Ali Abdourahmane

Student ID: 30592027

GitHub Repository Link: <https://github.com/AbdourahmaneFatmaAli/ai-course-lab0.git>

Date Submitted: 05/06/2026

1. AI Use Summary

Question	Student Response
Did you use any AI tool for this lab?	YES
Estimated percentage of the work influenced by AI	30%
Did you attach evidence of AI use where applicable?	YES

2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

Name of AI Tool Used	Purpose for Using the Tool	Prompt or Instruction Given to the Tool	Part(s) of the Work Influenced by the Tool	How I Verified, Edited, or Improved the AI Output
ChatGPT	To understand some part of the requirements and to debug	Explain the lab, OR how to fix this error	Importing Numpy, Pandas, Matplotlib. With vs code setup	AI sometimes gives what is not required so its important to keep checking whether we are doing the right thing

3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	Attached? (Yes, No, N/A)	File Name
AI-generated draft		
Prompt history		

Screenshot of AI interaction	YES	
Exported AI conversation		
AI-generated code snippet	YES	
Revised version showing student input		

4. Student Declaration

I declare that:

Declaration Statement	Response (Yes/No)
The submitted work is my own work.	
Any use of AI tools has been clearly declared in this form.	YES
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	YES
I reviewed, tested, edited, and improved any AI-generated content before submission.	YES
My AI usage does not exceed 25% of the entire work.	NO
I understand that undeclared or excessive AI use may be treated as academic misconduct.	YES

Student Signature: _Fatma Ali

Abdourahmane_____

Date: _____ 05/06/2027 _____

AI-generated code snippet:

git add .

git commit -m "Initial Lab 0 submission"

git commit -m "Complete NumPy and Pandas exercises"

git remote add origin https://github.com/YOUR_USERNAME/ai-course-lab0.git

git branch -M main

git push -u origin main

Nouveau chat

Bibliothèque

Projets

Applications

Codex

Plus

Récents

Lab 0 Environment Setup

Mesure de tension artérielle

Search Algorithms Overview

Agent Types Explanation

Surdosage d'Effergan

Fatma Ali Abdourahmane

videos of tutorial on this subject

These are some of the best beginner-friendly video tutorials for exactly the topics in your lab:

Full Beginner Courses (Best Overall)

1. freeCodeCamp — Data Analysis with Python

Covers:

NumPy

Pandas

Matplotlib

Seaborn

JUST EXPLAIN the lab instructions

Here's a clear explanation of what this Lab 0 is actually asking you to do, without the extra technical noise.

What this Lab is about (big picture)

This lab is not about advanced AI yet. It is checking if you can:

Set up your computer for programming

Use Python correctly

Use Git + GitHub for saving code online

Use Google Colab (cloud notebooks)

Do basic data work with NumPy, Pandas and plotting

Think of it as:

You already completed a big part of Part 1 earlier. Based on your previous messages, you have:

- Installed Python
- Created the `ai_env` virtual environment
- Opened the project in VS Code
- Opened `python_basics.ipynb`
- Selected a Python interpreter/environment

So now your main goal is to finish the required deliverables for submission.

Here's the fastest path to complete Lab 0.

1. Finish Exercise 1 — Environment Verification

Create a file named:

`environment_test.py`



ChatGPT

Partager

```
(ai_env) PS C:\Users\Fatma Ali\ai_labs\ai-course-lab0>
python environment_test.py
Python: 3.13.7
OS: Windows
numpy: 2.4.6
pandas: 3.0.3
matplotlib: 3.10.9
seaborn: 0.13.2
(ai_env) PS C:\Users\Fatma Ali\ai_labs\ai-course-lab0>
Afficher moins ^
```

ChatGPT ▾

↑ Partager

● Problem in your VS Code right now

You selected:

✗ Python 3.13 (global Python)

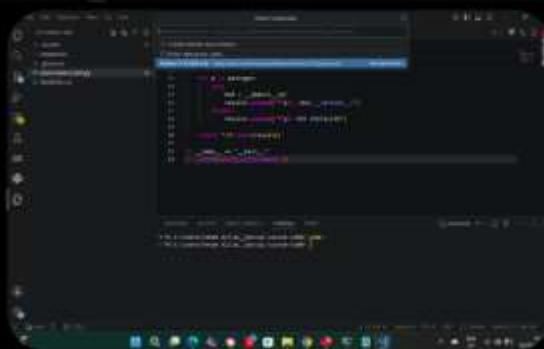
NOT your virtual environment:

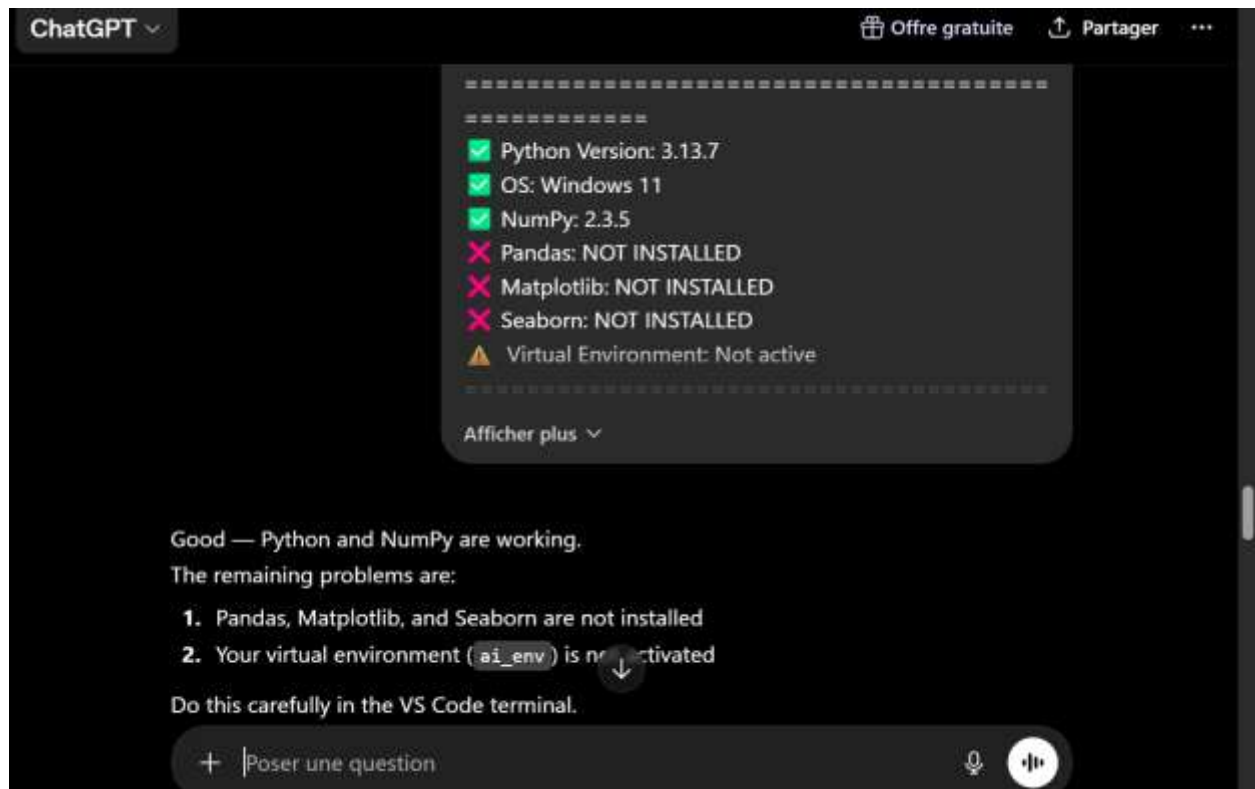
✗ ai_env is NOT selected

That's why VS Code doesn't show `ai_env` and your lab environment feels "missing".

ChatGPT ▾

↑ Partager ...





Done by student

```
plt.figure(figsize=(8,5))
```

```
sns.scatterplot(
    x='hours_studied',
    y='exam_score',
    hue='major',
    data=df
)
```

```
sns.regplot(
```

```
x='hours_studied',
y='exam_score',
data=df,
scatter=False,
color='black'
)

plt.title("Hours Studied vs Exam Score")
plt.show()
```

Edited :

```
import pandas as pd
```

```
import numpy as np
```

```
np.random.seed(42)
```

```
n_students = 200
```

```
data = {
    'student_id': range(1000, 1000 + n_students),
    'major': np.random.choice(['CS', 'Math', 'Physics', 'Biology'], n_students),
    'year': np.random.choice([1, 2, 3, 4], n_students),
    'exam_score': np.random.normal(75, 10, n_students).clip(0, 100),
    'assignments_completed': np.random.randint(0, 11, n_students),
    'hours_studied': np.random.normal(15, 5, n_students).clip(1, 40)
}
```

```
df = pd.DataFrame(data)
```

Edited

Normalization

```
np.random.seed(42)
random_data = np.random.randn(100, 3)

normalized = (random_data - random_data.mean(axis=0)) / random_data.std(axis=0)

normalized[:5]
```

[2] Python

```
array([[ 0.49343924,  0.04616031,  0.51734005],
       [ 1.74402998, -0.05226901, -0.27901188],
       [ 1.81249029,  0.97585261, -0.49153853],
       [ 0.54930358, -0.28760687, -0.48815686],
       [ 0.18301785, -1.77587837, -1.62529439]])
```

Linear regression

```
X = np.random.randn(50, 3)
true_theta = np.array([2.5, -1.2, 3.7])
y = X @ true_theta + np.random.randn(50) * 0.1

theta_hat = np.linalg.inv(X.T @ X) @ X.T @ y
```

[3] Python

Cell 3 of 50

Correlation

```
print(df[['hours_studied', 'exam_score']].corr())
```

Python

	hours_studied	exam_score
hours_studied	1.000000	-0.036509
exam_score	-0.036509	1.000000

performance categories

```
df['performance'] = pd.cut(
    df['exam_score'],
    bins=[0, 70, 80, 90, 100],
    labels=['Needs Improvement', 'Average', 'Good', 'Excellent']
)

df.head()
```

Python

	student_id	major	year	exam_score	assignments_completed	hours_studied	performance
0	1000	Physics	3	68.199753	10	10.922505	Needs Improvement
1	1001	Biology	4	77.322537	9	12.204873	Average

Group by major + year

```
df.groupby(['major', 'year']).agg({  
    'student_id': 'count',  
    'exam_score': 'mean',  
    'hours_studied': 'mean'  
})
```

[13]

		student_id	exam_score	hours_studied
major	year			
Biology	1	13	76.570059	12.677806
	2	12	74.454155	13.426372
	3	14	80.595336	15.095221
	4	15	70.697689	14.253944
CS	1	12	76.716704	13.039480
	2	7	76.155936	13.595825
	3	9	72.347626	15.039892
	4	18	78.119997	16.275541
Math	1	11	74.554794	19.001215
	2	9	82.669539	12.654613
	3	10	72.017537	15.175871
	4	16	72.102066	17.217642

Done by student and fixed by AI



Mistake: forgot to create and use subplots (

`plt.figure(figsize=(12, 5))`

`sns.histplot(df['exam_score'], kde=True)`

`sns.boxplot(x='major', y='exam_score', data=df)`

and fixed by AI

`fig, axes = plt.subplots(1, 2, figsize=(12, 5))`

`# Histogram + KDE`

`sns.histplot(df['exam_score'], kde=True, ax=axes[0])`

`axes[0].set_title("Exam Score Distribution")`

`# Boxplot by major`

`sns.boxplot(x='major', y='exam_score', data=df, ax=axes[1])`

```
axes[1].set_title("Scores by Major")
```

```
plt.tight_layout()
```

```
plt.show()
```

Reference of videos watched:

<https://www.youtube.com/watch?v=XFZRVnP-MTU>

<https://www.youtube.com/watch?v=4DnWYK88-E4>

<https://www.youtube.com/watch?v=UO98lJQ3QGI>

<https://www.youtube.com/watch?v=Hb3RmWowIKw>